Milestone-2

Objective
In this phase, you will analyze how disaster declarations evolve over time. The goal is not just to create plots, but to identify patterns, trends, and seasonality in the data.
As defined in the project workflow
visualizing US Natural disaster
, this stage focuses on:

- Temporal trends
- Trend comparison across incident types

◈ STEP 1: Validate Time Data

Before analysis, ensure your time-related data is correct.
Checklist:

- Confirm date column is in datetime format
- Ensure extracted columns exist:
  - year
  - month
- Check for missing or invalid dates

Important:
If date is incorrect → **all further analysis becomes invalid**

◈ STEP 2: Yearly Trend Analysis

Understand how total disasters change over time
Tasks:

- Group data by year
- Count number of disaster declarations per year

Visualization:

- Line chart:
  - X-axis → Year
  - Y-axis → Number of disasters

Interpretation:
Justify for:

- Is there an increasing trend?
- Any sudden spikes?
- Any abnormal drops?

◈ STEP 3: Incident Type Trends Over Time

Compare how different disaster types evolve over time

Tasks:

- Group data by:
  - year + incidentType
- Aggregate counts

Visualization:

- Multi-line chart:
  - X-axis → Year
  - Y-axis → Count
  - Lines → Different incident types

Interpretation:

- Which disaster type is most frequent?
- Which type is increasing fastest?
- Any type declining?

◈ STEP 4: Seasonality Analysis (Monthly Trends)

Identify patterns within a year

Tasks:

- Group data by month
- Count disasters per month

Visualization:

- Line chart OR bar chart:
  - X-axis → Month
  - Y-axis → Count

Interpretation:

- Which months have highest disasters?
- Are disasters seasonal?
- Any recurring pattern?

◈ STEP 5: Advanced Time Insights (Optional but Recommended)

◈ 5.1 Rolling Average

- Apply moving average (window = 3 or 5 years)
- Helps smooth noise

◈ 5.2 Growth Rate

- Calculate year-over-year percentage change

◈ 5.3 Peak Detection

- Identify years with highest disasters

◈ STEP 6: Visualization Standards (MANDATORY)

Every plot must follow:

Required Elements:

- Title
- X-axis label
- Y-axis label
- Legend (if multiple lines)

Best Practices:

- Avoid clutter
- Use readable fonts
- Ensure proper scaling

Things not to do:

- Use random colors
- Plot without labels
- Overload graph with too many lines

◈ STEP 7: Insight Writing

For **each visualization**, write:

Format:

1. What does the graph show?
2. What pattern do you observe?
3. What could be the reason?

Documents to be submitted:

1. Milestone-2 document (in pdf format)
2. Jupyter notebook code file (in .ipynb)